

Prof. dr. Kirsten ten Tusscher



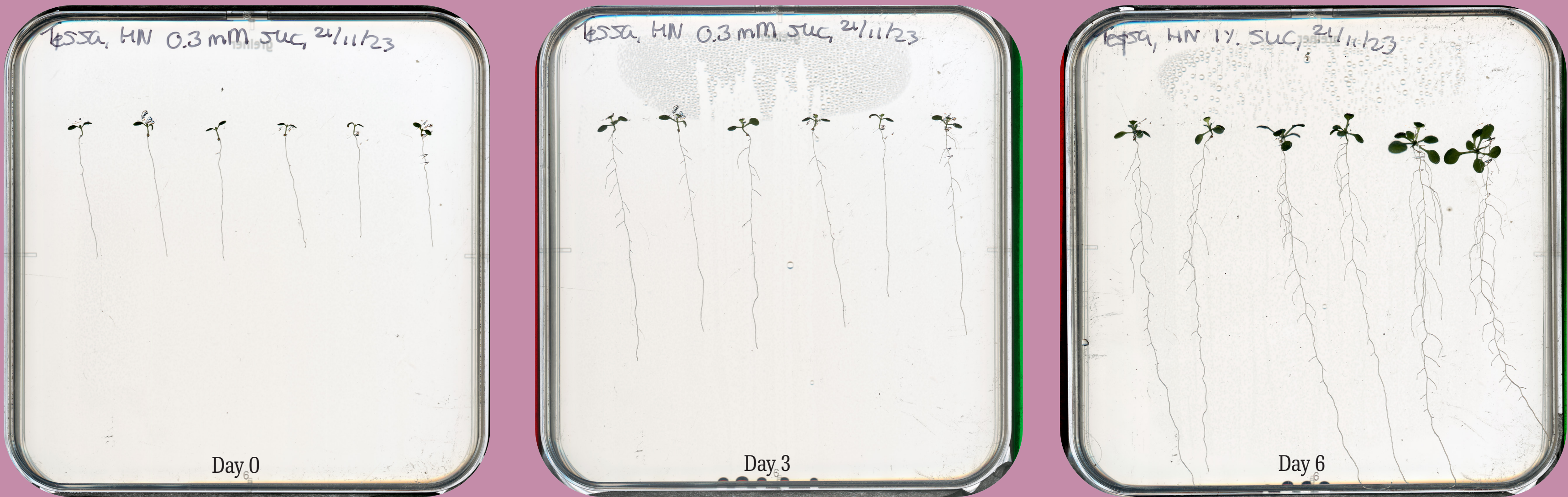
I study how roots from *Arabidopsis thaliana*, but also plants like tomato and potato keep growing throughout their life, from how they form side branches, to how they adapt to the conditions in which they live. Based on this information from experiments, we also create computational models to understand how a plant processes information and “decides” what to do.

ON THE IMPACT

I like to show through my research that plants, which don't have a brain, can make smart decisions. Plants can “decide” when and where to invest in growth based on the information they can sense about their own condition and that of their environment, as if they were doing investment banking.

There are other examples; some computer programs are inspired by how plants process information. Plant research has taught us so much and continues to do so.

Can we make
plant research
accessible to all
publics?



Arabidopsis plants growing on plates with nutrients for 6 days; notice the different sizes of the plants over time (by Lucila Salvatore). We need to lable each sample to keep track of which sample is which. We lable with our names, the date and the conditions.

ON PREVIOUS EXPERIENCES

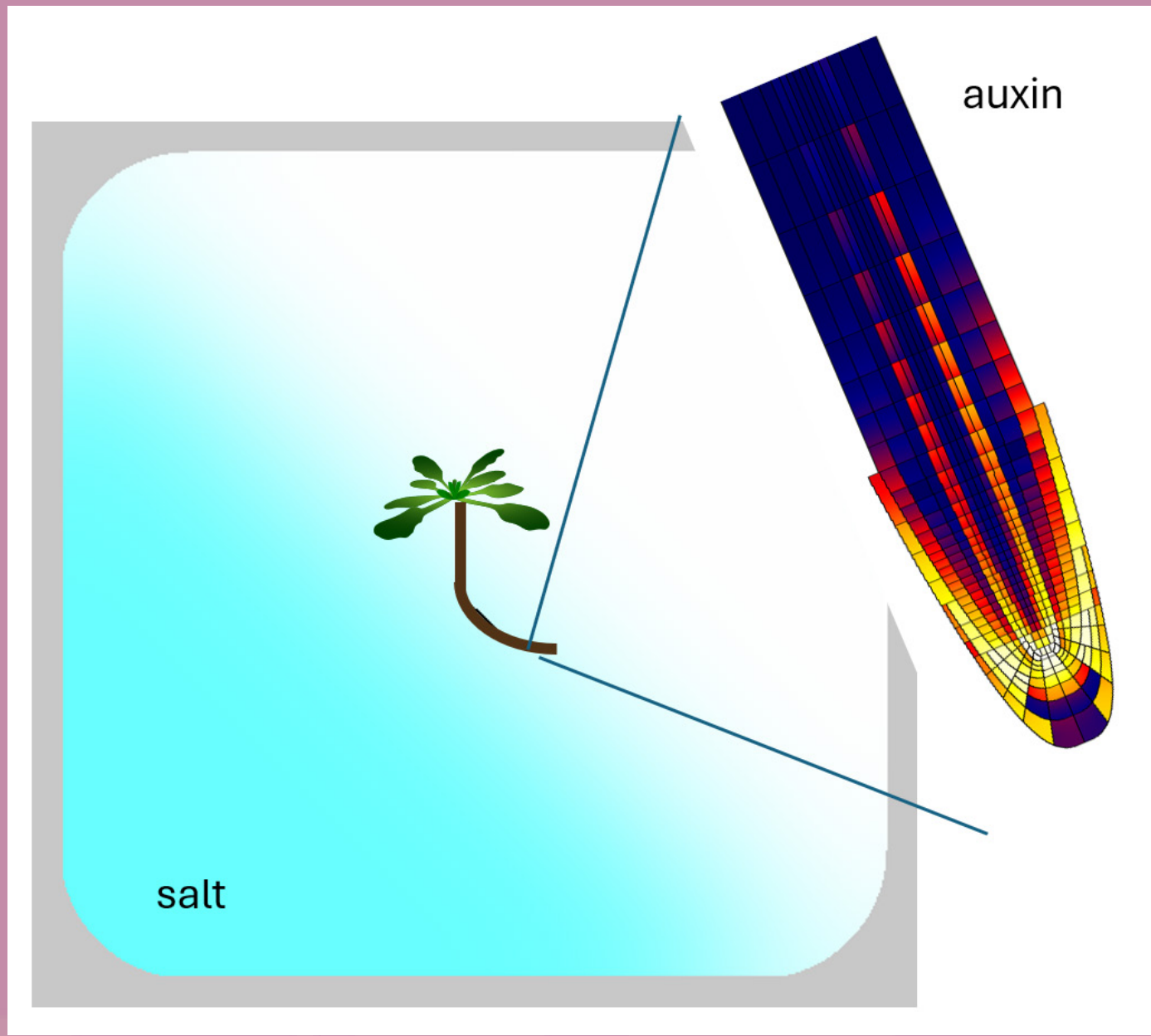
Me and one of my PhD students went to primary schools to share our research, and that was very nice. I also remember giving a talk to my son's class about my research. In both situations, getting the kids excited and them talking to their own parents about it helped me realize we can reach multiple generations at once.

The outreach activities that I do are mostly visual and interactive since it is easier to explain my research in this way. I once showed a group of schoolchildren how a plant grown in a maze is able to change its direction of growth when it encounters physical (or environmental) restraints.

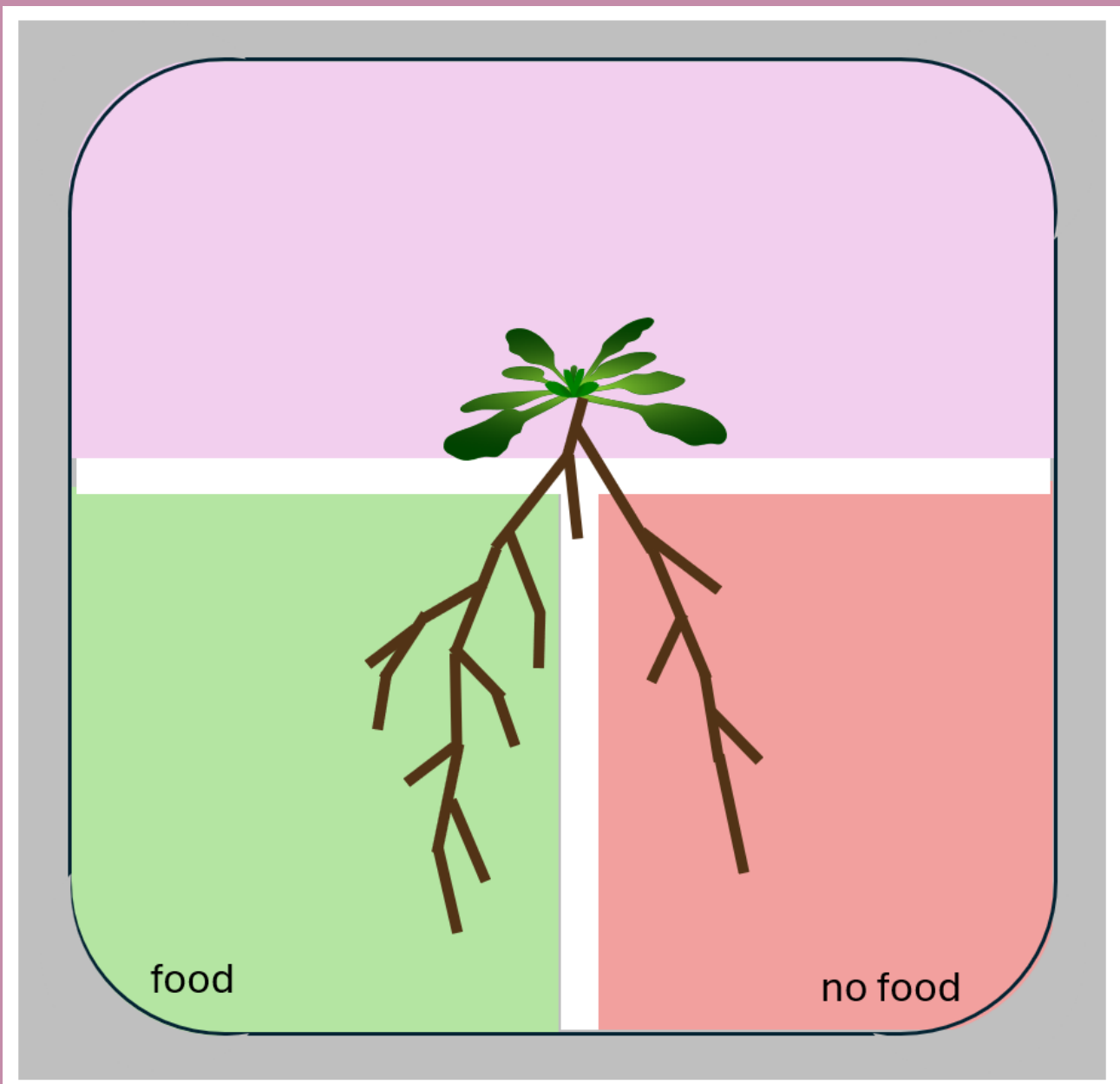
ON THE CHALLENGES

In collaborations with colleagues who do experimental work, it can be challenging to translate some of the computational nature of my work, and vice versa. The challenge continues when translating my work to the general public. That is why I like to find analogies that connect every-day things to my research.

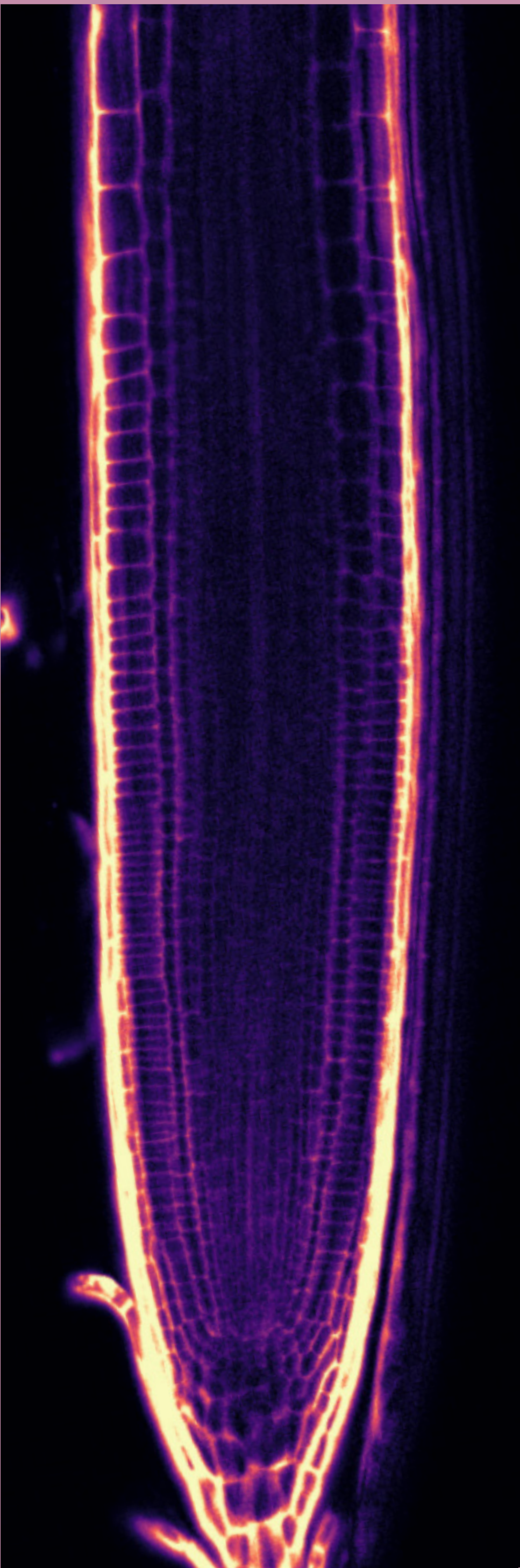
Furthermore, having patience while the gap between our fundamental research and applications closes is also challenging. Increasingly, we see modelling and systems thinking in mainstream education, but it has taken a few years.



These are some examples of how we can show our research in a visual way. Plants can sense their environment. They can grow away from salt stress (left side) or grow more in areas where they sense more food. We use computer models to identify genes responsible for this behavior.



This is an example of what a buildable game could look like for a plant with realistic roots (image created with ChatGPT).



Arabidopsis thaliana root with a cell wall stain (by Naomi Lubeeek).

ON THE FUTURE

I think there are a few things we can do as scientists for science communication:

Focus on engaging themes, like how plants make decisions based on their environment.

Take as many opportunities as possible to communicate our research in approachable ways. We can reach more groups than we think.

Develop interactive, visual tools (e.g. realistic root models) with multidisciplinary teams to showcase the creativity and complexity of plant science and of the plants themselves!

Read this poster in Dutch



What do you think?

